CSE370 Spring 2026
Database Systems
Assignment: **01**
No of Questions: **03**
Total Marks: 20

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
2. Your answer should be **handwritten**
3. Submission Form: https://forms.gle/9aTMkVyfwhLfn1aE8
4. **NO LATE SUBMISSION WILL BE ACCEPTED.**

Q1. [2+2 Marks] Select if the following tasks will require DDL(Data Definition Language) or DML(Data Manipulation Language).
Explain your answer briefly (at max 2 sentences) with mention to schema and state:

a. Rename the column called "StudentID" into "SID"
Ans: Renaming the column modifies the schema of the database. DDL is applicable in this case because it affects how the data is defined but not the actual data.
b. Change the "class time" for section 7 to "01:00 PM - 2:05 PM"
Ans: Changing the class time updates the state stored in the table. DML is applicable in this case because it changes the content of existing records but not the schema.

Q2. [3+3 Marks] There are two types of data independence in database management systems: Logical Data Independence and Physical Data Independence. Given the scenarios below, **identify** what type of **data independence** both are and **explain your answer briefly** with reference to the **Three-Schema Architecture**.

- You work for a social media company. The company has decided to move from cloud hosting to physical hardware to support the server due to increase in demand.
Ans: Physical Data Independence.
Changes Internal Schema without affecting Conceptual Schema.
- You work for a social media company. The company has decided to track the number of hours a user has stayed on their platform. This tracked_hours is not visible to the user but kept in the database.
Ans: Logical Data Independence.
Changes Conceptual Schema without affecting External Schema.

Q3. [10 Marks] You are tasked with designing a Video Streaming System. The system must meet the following requirements:

- Each user will have a unique code, full name, email, and mobile number.
- Users can watch videos, which have a unique video ID, duration, upload_date, and play count.
- A video can be watched by many users and a user can watch many videos.
- Users can also comment on videos. Whenever a comment is made, the timestamp is recorded. A single user can comment multiple times.
- All videos must have comments, but not all users will leave a comment on the videos.
- Users can also follow other users.

- Some users might create a playlist. Each playlist has a playlist_name, tag, and a list of videos. Playlist_name is not unique, rather it is kept track of with the user's username and playlist_name combined.
- Users add videos to their playlist. One user can add many videos to their single playlist.

Now, draw an **ER diagram** of this system. Write down all assumptions (if any). It is mandatory to show the **cardinality ratio** and the **participation type** for each relationship.